

EX NO:2	Chebyshev IIR Filter
DATE:	2.2. Design and Performance Evaluation of a Chebyshev Type-I IIR High-Pass Filter

```

clc;
clear;
close all;
% Specifications
Rp = 1; % Passband Ripple (dB)
Rs = 40; % Stopband Attenuation (dB)
Wp = 0.5; % Passband Edge
Ws = 0.35; % Stopband Edge
% Find Filter Order
[n,Wn] = cheb1ord(Wp,Ws,Rp,Rs);
% Design High-Pass Filter
[b,a] = cheby1(n,Rp,Wn,'high');
% Frequency Response
figure;
freqz(b,a);
title('Frequency Response');
% Impulse Response
figure;
impz(b,a);
title('Impulse Response');
% Group Delay
figure;
grpdelay(b,a);
title('Group Delay');
% Pole-Zero Plot
figure;

```

```
zplane(b,a);
```

```
title('Pole-Zero Plot');
```

```
fprintf('Filter Order = %d\n',n);
```

```
fprintf('Cutoff Frequency = %.4f\n',Wn);
```



